

CASE STUDY 3: Library Book Management System

```
// Create Database
```

```
use LibraryDB
```

```
// Create Collection
```

```
db.createCollection("books")
```

```
// Insert One Book Record
```

```
db.books.insertOne({  
    bookId: 1,  
    title: "MongoDB Basics",  
    author: "Tom",  
    price: 450,  
    category: "Database"  
})
```

```
// Insert Multiple Book Records
```

```
db.books.insertMany([  
    {  
        bookId: 2,  
        title: "Java Fundamentals",
```

```
    author: "James",  
    price: 500,  
    category: "Programming"  
  },  
  {  
    bookId: 3,  
    title: "Python Guide",  
    author: "Guido",  
    price: 480,  
    category: "Programming"  
  },  
  {  
    bookId: 4,  
    title: "Data Science 101",  
    author: "Andrew",  
    price: 700,  
    category: "Analytics"  
  },  
  {  
    bookId: 5,  
    title: "Web Development",  
    author: "Mark",
```

```
    price: 550,  
    category: "Web"  
  },  
  {  
    bookId: 6,  
    title: "AI Basics",  
    author: "Alan",  
    price: 800,  
    category: "AI"  
  },  
  {  
    bookId: 7,  
    title: "Cloud Computing",  
    author: "Jeff",  
    price: 650,  
    category: "Cloud"  
  },  
  {  
    bookId: 8,  
    title: "Cyber Security",  
    author: "Kevin",  
    price: 600,
```

```
    category: "Security"
  },
  {
    bookId: 9,
    title: "Big Data",
    author: "Hadoop",
    price: 750,
    category: "Analytics"
  },
  {
    bookId: 10,
    title: "SQL Mastery",
    author: "Oracle",
    price: 400,
    category: "Database"
  }
])
```

```
// Handle Duplicate Book IDs
```

```
// Create Unique Index
```

```
db.books.createIndex({bookId:1}, {unique:true})
```

```
// Bulk Insert with Duplicate Handling
```

```
db.books.insertMany([
```

```
{
```

```
  bookId: 11,
```

```
  title: "Machine Learning",
```

```
  author: "John",
```

```
  price: 900,
```

```
  category: "AI"
```

```
},
```

```
{
```

```
  bookId: 1,
```

```
  title: "Duplicate Book",
```

```
  author: "Test",
```

```
  price: 300,
```

```
  category: "Database"
```

```
}
```

```
], {ordered:false})
```

```
// ordered:false allows other records to insert
```

```
// even if one duplicate exists
```

```
// Retrieve All Book Records
```

```
db.books.find()
```

```
// Retrieve One Book Record by Book ID
```

```
db.books.findOne({bookId:3})
```

```
// Filter Books Based on Category
```

```
db.books.find({category:"Programming"})
```

```
// Filter Books Based on Author
```

```
db.books.find({author:"Tom"})
```

```
// Filter Books Based on Price Range
```

```
db.books.find({  
  price: {$gte:500, $lte:800}  
})
```

```
// Display Only Selected Fields
```

```
db.books.find(  
  {},  
  {  
    _id:0,
```

```
        title:1,  
        price:1  
    }  
)
```

```
// Remove Collection  
db.books.drop()
```

```
// Remove Database  
use LibraryDB  
db.dropDatabase()
```

OUTPUT:

```

> use LibraryDB
< switched to db LibraryDB
> db["books"].find()
<
> show collections
< books
> db.books.insertOne({
  bookId: 1,
  title: "MongoDB Basics",
  author: "Tom",
  price: 450,
  category: "Database"
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a100f8d841fc0f4efb9760c')
}

```

```

  category: "Database"
}
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a100fc5841fc0f4efb9760d'),
    '1': ObjectId('6a100fc5841fc0f4efb9760e'),
    '2': ObjectId('6a100fc5841fc0f4efb9760f'),
    '3': ObjectId('6a100fc5841fc0f4efb97610'),
    '4': ObjectId('6a100fc5841fc0f4efb97611'),
    '5': ObjectId('6a100fc5841fc0f4efb97612'),
    '6': ObjectId('6a100fc5841fc0f4efb97613'),
    '7': ObjectId('6a100fc5841fc0f4efb97614'),
    '8': ObjectId('6a100fc5841fc0f4efb97615')
  }
}

```


> MONGOSH

```
> db.books.createIndex({bookId:1}, {unique:true})
```

```
< bookId_1
```

```
> db.books.insertMany([
```

```
{
```

```
  bookId: 11,
```

```
  title: "Machine Learning",
```

```
  author: "John",
```

```
  price: 900,
```

```
  category: "AI"
```

```
},
```

```
{
```

```
  bookId: 1,
```

```
  title: "Duplicate Book",
```

```
  author: "Test",
```

```
  price: 300,
```

```
  category: "Database"
```

```
}
```

```
], {ordered:false})
```

```
], {ordered:false})
```

```
> MongoBulkWriteError: E11000 duplicate key error collection: LibraryDB.books index: bookId_1 dup key: { bookId: 1 }
```

```
db.books.find()
```

```
{
```

```
  _id: ObjectId('6a100f8d841fc0f4efb9760c'),
```

```
  bookId: 1,
```

```
  title: 'MongoDB Basics',
```

```
  author: 'Tom',
```

```
  price: 450,
```

```
  category: 'Database'
```

```
}
```

```
{
```

```
  _id: ObjectId('6a100fc5841fc0f4efb9760d'),
```

```
  bookId: 2,
```

```
  title: 'Java Fundamentals',
```

```
  author: 'James',
```

```
  price: 500,
```

Activate Windows

```
> db.books.findOne({bookId:3})
< {
  _id: ObjectId('6a100fc5841fc0f4efb9760e'),
  bookId: 3,
  title: 'Python Guide',
  author: 'Guido',
  price: 480,
  category: 'Programming'
}
> db.books.find({category:"Programming"})
< {
  _id: ObjectId('6a100fc5841fc0f4efb9760d'),
  bookId: 2,
  title: 'Java Fundamentals',
  author: 'James',
  price: 500,
  category: 'Programming'
}
```

```
db.books.find({author:"Tom"})
{
  _id: ObjectId('6a100f8d841fc0f4efb9760c'),
  bookId: 1,
  title: 'MongoDB Basics',
  author: 'Tom',
  price: 450,
  category: 'Database'
}
db.books.find({price:{$gte:500,$lte:800}})
{
  _id: ObjectId('6a100fc5841fc0f4efb9760d'),
  bookId: 2,
  title: 'Java Fundamentals',
  author: 'James',
  price: 500,
  category: 'Programming'
}
```

```
}
db.books.find({},{_id:0,title:1,price:1})
{
  title: 'MongoDB Basics',
  price: 450
}
{
  title: 'Java Fundamentals',
  price: 500
}
{
  title: 'Python Guide',
  price: 480
}
{
  title: 'Data Science 101',
  price: 700
}
```

```
    price: 500  
  }  
> db.books.drop()  
< true  
> db.dropDatabase()  
< { ok: 1, dropped: 'LibraryDB' }
```